# Alyssa Wu

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## EDUCATION

**Pomona College — Bachelor of Arts** *Expected May 2028*
Computer Science & Public Policy Analysis (Science, Technology, Society) Double Major, Math Minor GPA: 3.98
Notable Courses: Human Centered Computing, Computational Design Tools, Data Structures, Artificial & Bio Vision, Space & GIS

## EXPERIENCE

**The Hive (The Rick and Susan Sontag Center for Collaborative Creativity) — Data Analyst** *Jan 2025 - Present*
- Redesigning space-usage, attendance, and user-experience tracking pipelines using human-centered design methods, conducting needs-finding interviews, contextual inquiry, and prototype testing with students and staff
- Designing and building The Hive's first data dashboard and custom CRM to inform event programming, manage alumni outreach, and communicate student impact to donors and consortium presidents

**Microsoft — Undergraduate Research Intern** *May 2026 - Aug 2026*
- Evaluated 5+ machine learning architectures for county-level power outage prediction using projected hurricane paths;; visualized predictions against ground truth across 10 U.S. hurricanes on custom GUI
- Constructed leakage-free, multi-timestep forecast and reanalysis datasets to explore feature richness of Aurora weather-model embeddings, storm features, and socioeconomic features
- Developed MCP-based skills for a power system agent, integrating scheduled outages with traceable powerflow analysis, crew dispatch, and belief state updates visualized on a geospatial GUI

**The Nature Conservancy — Artificial Intelligence Software Engineer Intern** *May 2025 - July 2025*
- Built SPECTER-based NLP model with 0.96 recall in classifying agroforestry literature relevance from titles and abstracts
- Ran comparative experiments on 15+ embedding and neural architecture combos with scikit-learn, PyTorch, and Hugging Face
- Built and delivered deployable Google Colab for Tackle Climate Change team to streamline agroforestry evidence synthesis

**Newark Office of U.S. Senator Cory Booker — Intern** *Sept 2023 - Jun 2024*
- Attended to, classified, and tracked 100+ daily calls and voicemails for constituent database and casework data visualization
- Represented Senator at site visits on transit, e.g. Port Newark hybridization, Gateway Tunnel Project, and NY/NJ ferry subsidies

## PROJECTS

**Pinterest User Experience Research Study** *Fall 2026*
- Investigating how Pinterest can help users develop and articulate their personal taste through human-centered design research, including think-aloud sessions, diary studies, contextual inquiry, and Wizard-of-Oz prototyping

**Reel Hunters, Computational Design Tool** *Spring 2026*
- Iteratively designed, prototyped, and built a React Native mobile experience exploring how short-form video platforms can encourage intentional content consumption, drawing on interaction patterns from Pokémon GO and Instagram Reels

**Detecting Building Natural Disaster Vulnerability for American Red Cross** *Fall 2025 - Spring 2026*
- Collaborated with nonprofit to develop computer vision pipelines in Python and GIS to extract building vulnerability indicators from 360° street-view imagery for resilience mapping; experimented with YOLO, Faster R-CNN, and SphereNet

**P-AI Project Audio — Sampling Detection in Hip Hop with Deep Learning)** *Fall 2025*
- Investigated the robustness of audio fingerprinting with Mel Spectrograms and CNNs for automatic music sample identification, examining how machine representations capture perceptual similarities in remixed music

**Can I Get There Without A Car? — Claremont Consortium Accessibility Map** *Fall 2025*
- Designed participatory ArcGIS accessibility map for student to crowdsource and document local destinations, transportation modes, and barriers to car-free mobility

## SKILLS, CERTIFICATIONS, & INTERESTS

Python | Java | C | PyTorch | TensorFlow | Figma | ReactNative | ArcGIS | Git | Excel | Bloomberg Market Concepts Certified | Bloomberg Environmental Social Governance Certified | Cantonese | Mandarin | *Interests*: Earthen Urban Design, Piano, Fabrication

## AWARDS

**Princeton Prize in Race Relations (2024)** — Founded student-led BHM lessons (20+ lesson plans, 100+ classes taught) & organized free tutoring in underrepresented towns to combat academic segregation as president of Bergen County Academies' BLSU